

1

2

3

4

A

A

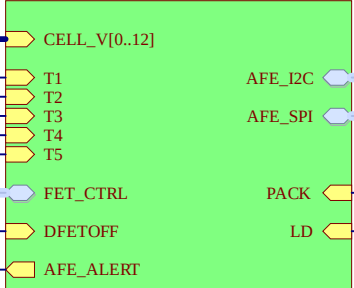
Battery Interface
Battery_Interface.SchDoc



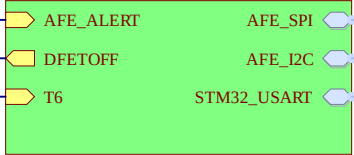
Temperature Sense
Temperature_Sense.SchDoc



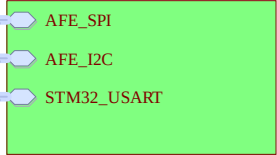
AFE
BQ76952_AFE.SchDoc



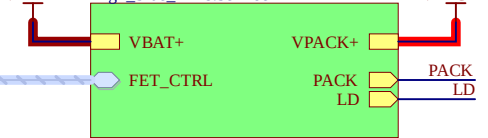
STM32 MCU
STM32_MCU.SchDoc



Pack Interface
Pack_Interface.SchDoc



High Side FETs
High_Side_FETs.SchDoc



Mounting Holes

MH1

MH2

MH3

MH4

1

2

3

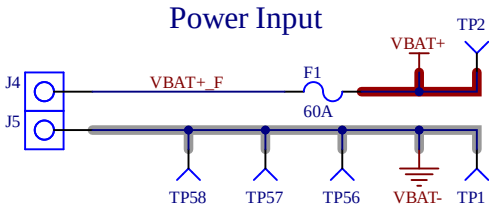
4

D

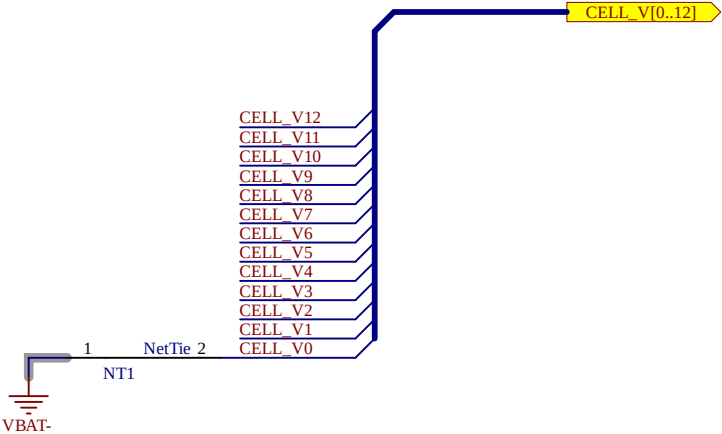
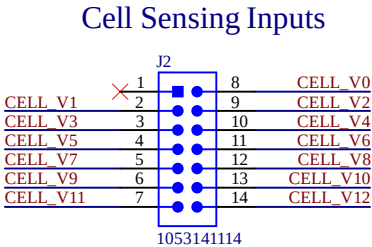
D

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PROJECT LTO 12S BMS.PrjPcb, [No Variations]		
DOCUMENT Top.SchDoc		MODIFIED 2026-05-18
ENGINEER Farris Matar	REVIEWER *	SHEET 1 OF 7

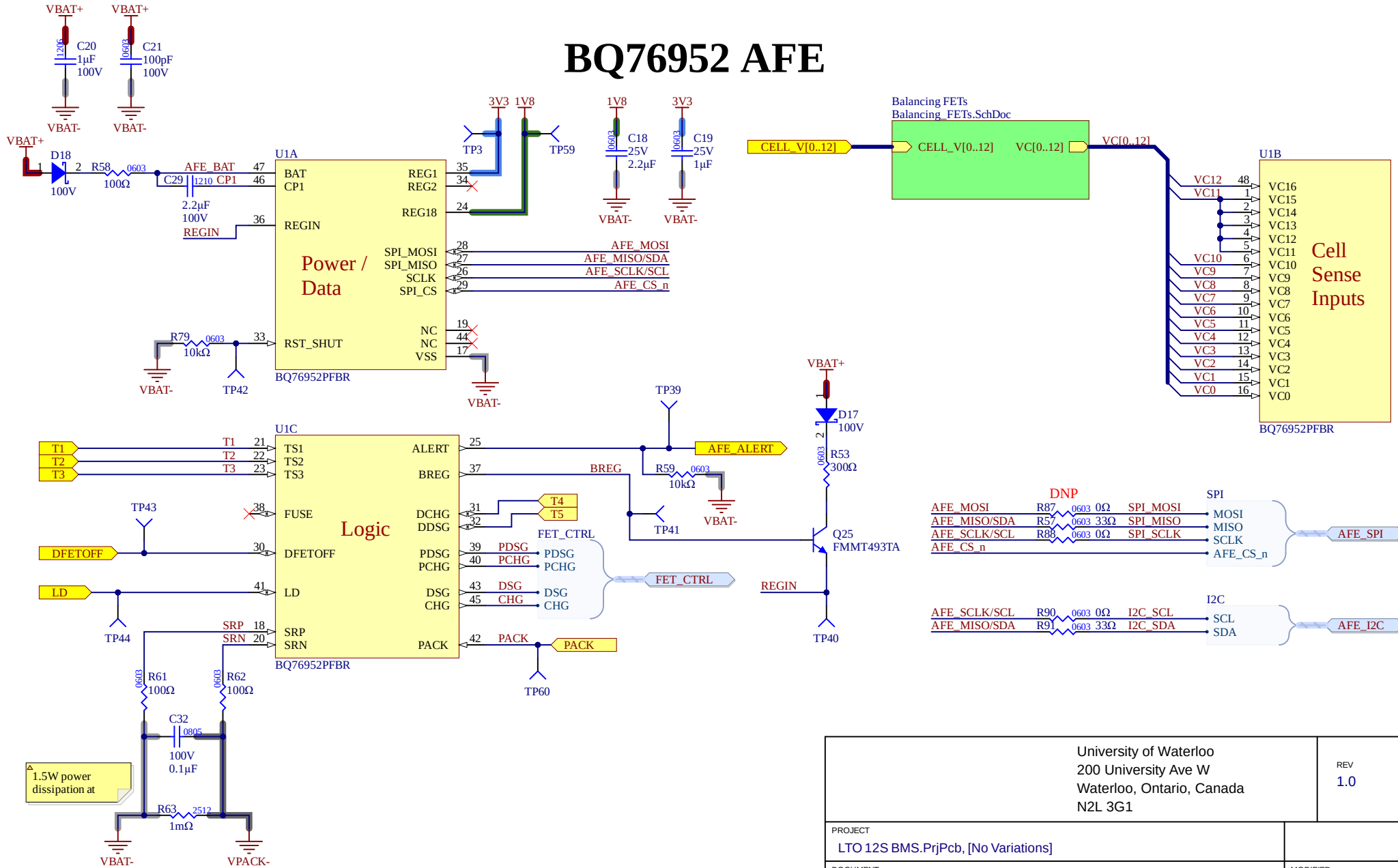
BATTERY INTERFACE



Battery specs:
- 12s1p 20Ah LTO battery
- 1.5V - 2.7V cell voltage range, 2.3V nominal



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DOCUMENT Battery_Interface.SchDoc		MODIFIED 2026-05-18
ENGINEER Farris Matar	REVIEWER *	SHEET 2 OF 7

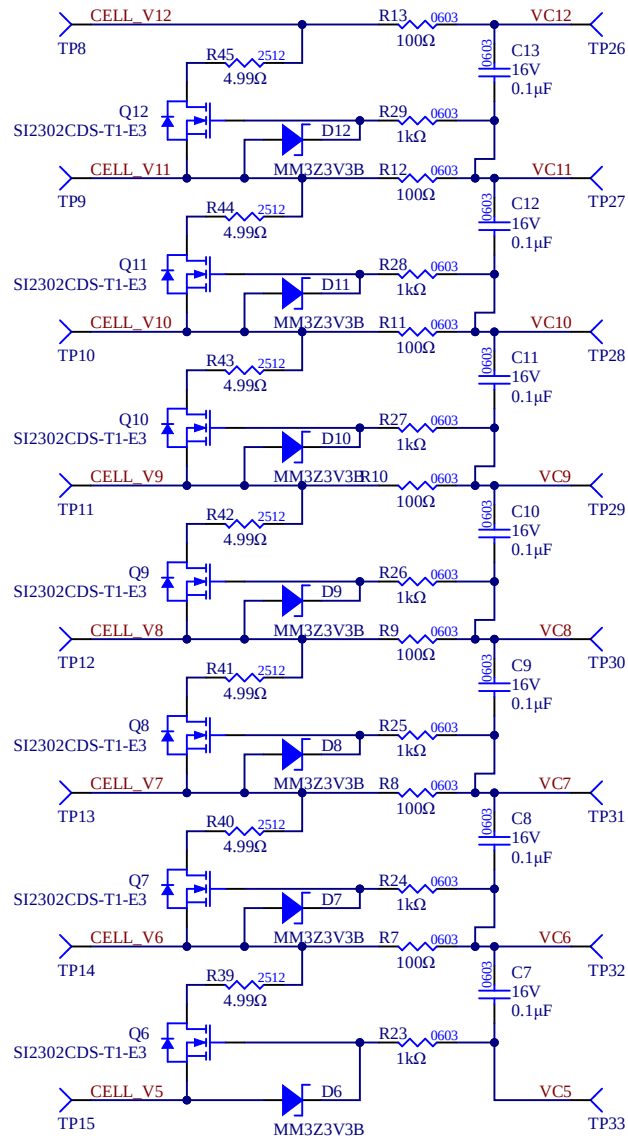


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PROJECT LTO 12S BMS.PrjPcb, [No Variations]		
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ENGINEER Farris Matar	REVIEWER *	SHEET 3 OF 7

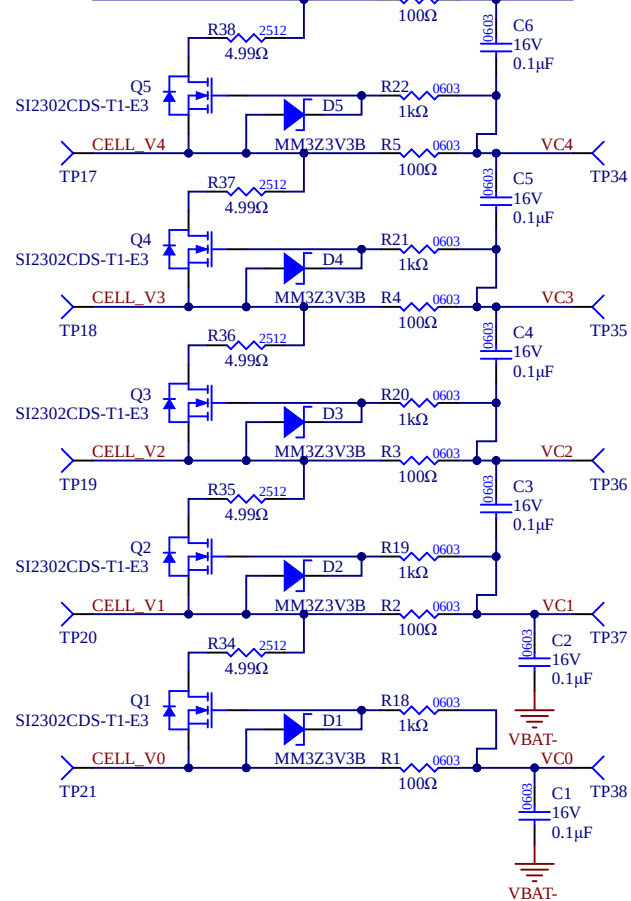
BALANCING FETS

CELL_V[0..12]

CELL_V12
CELL_V11
CELL_V10
CELL_V9
CELL_V8
CELL_V7
CELL_V6
CELL_V5
CELL_V4
CELL_V3
CELL_V2
CELL_V1
CELL_V0



CELL_V5



VC12
VC11
VC10
VC9
VC8
VC7
VC6
VC5
VC4
VC3
VC2
VC1
VC0

VC[0..12]

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REV
1.0

PROJECT
LTO 12S BMS.PrjPcb, [No Variations]

DOCUMENT
Balancing_FETs.SchDoc

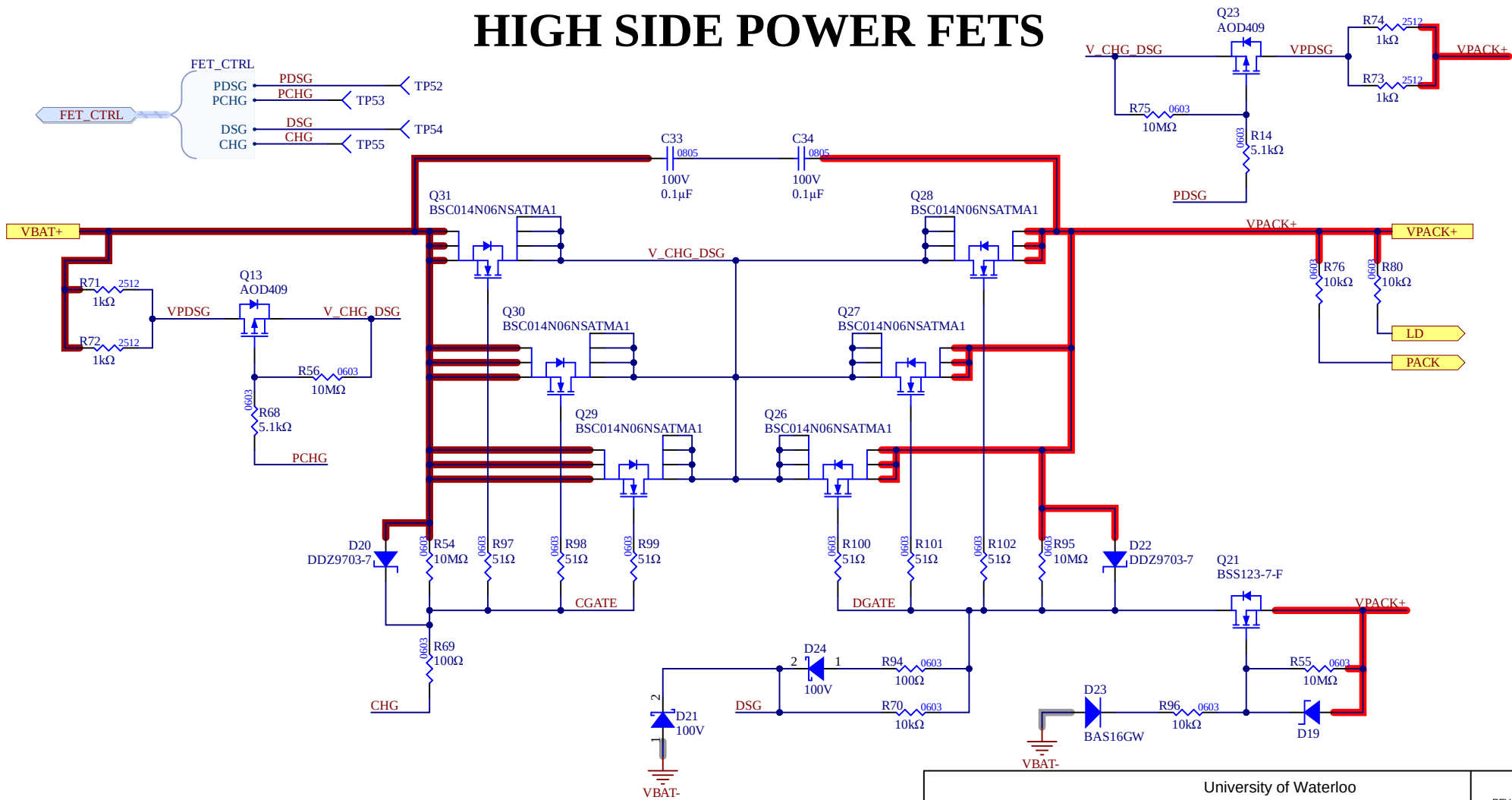
ENGINEER
Farris Matar

REVIEWER
*

MODIFIED
2026-05-18

SHEET 4 OF 7

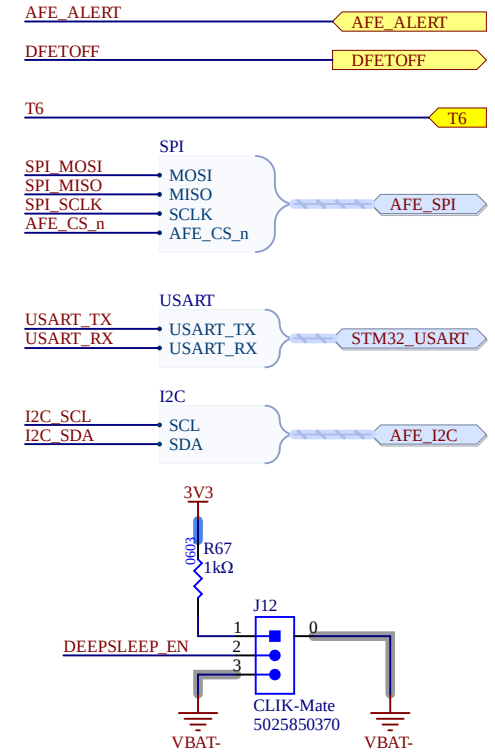
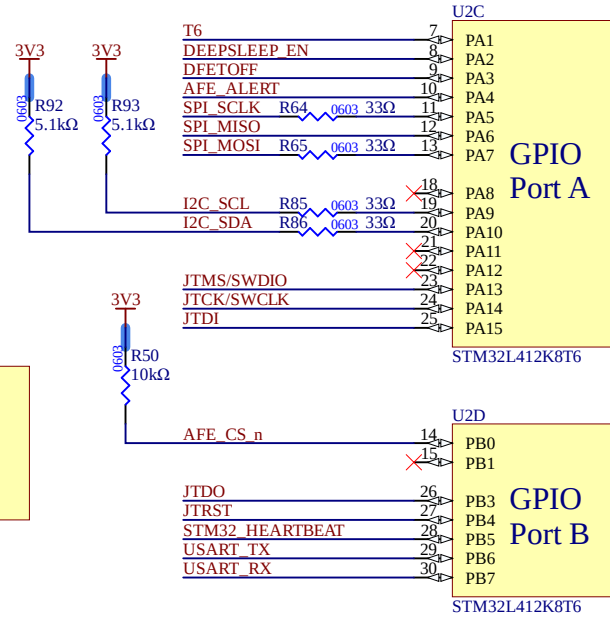
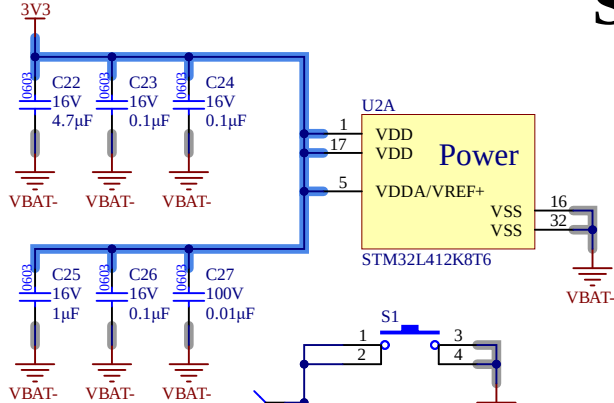
HIGH SIDE POWER FETS



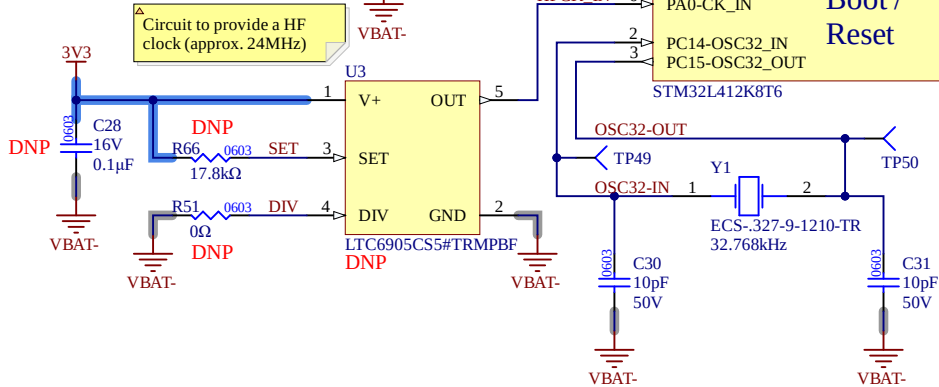
University of Waterloo 200 University Ave W Waterloo, Ontario, Canada N2L 3G1		REV 1.0
PROJECT LTO 12S BMS.PrjPcb, [No Variations]		
DOCUMENT High_Side_FETs.SchDoc		MODIFIED 2026-05-18
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STM32 MCU

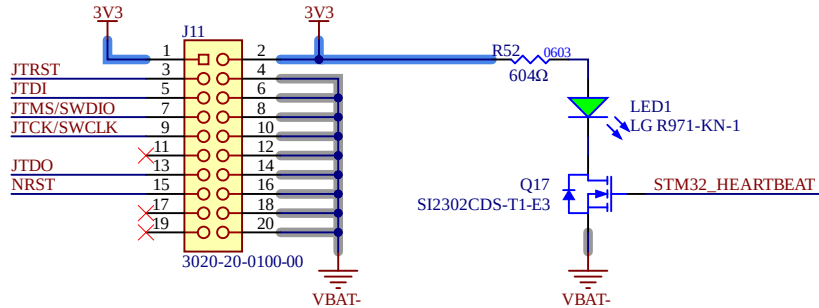
Follows reference design provided in
https://www.st.com/resource/en/application_note/an4555-getting-started-with-stm32l4-series-and-stm32l4-series-hardware-development-stmicroelectronics.pdf (Figure 14),
 minor modifications made to fit selected package



Connector to connect to a toggle switch that will trigger STM32 to put AFE in DEEPSLEEP (very low current consumption) and also put STM32 in SHUTDOWN mode until switch is flipped to wake it back up

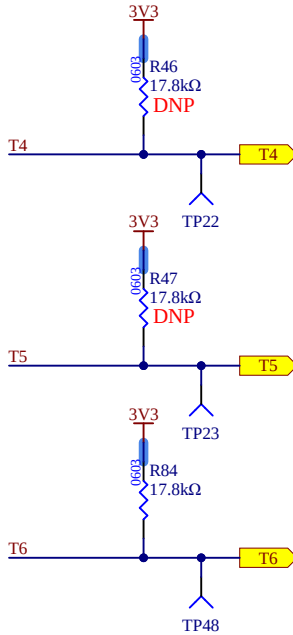
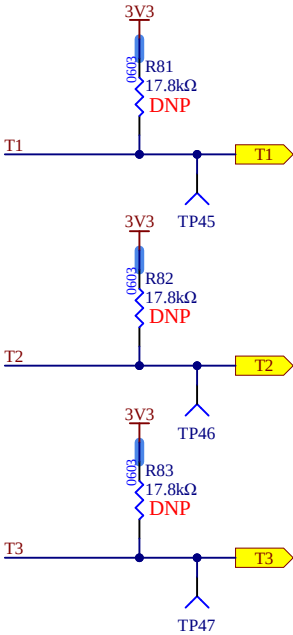
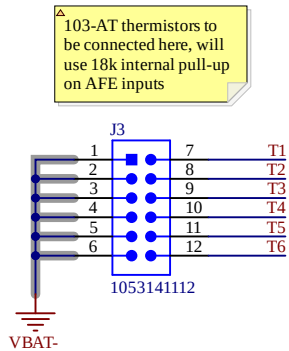


Debug / Programming Connector



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DOCUMENT STM32_MCU.SchDoc		MODIFIED 2026-05-18
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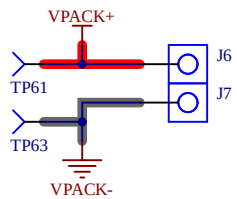
TEMPERATURE SENSING



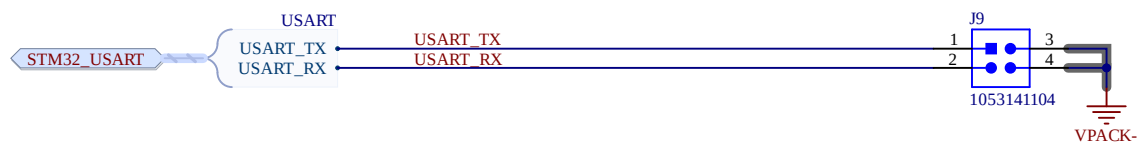
University of Waterloo 200 University Ave W Waterloo, Ontario, Canada N2L 3G1		REV 1.0
PROJECT LTO 12S BMS.PrjPcb, [No Variations]		
DOCUMENT Temperature_Sense.SchDoc		MODIFIED 2026-05-18
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PACK INTERFACE

Power Interface



Data Output

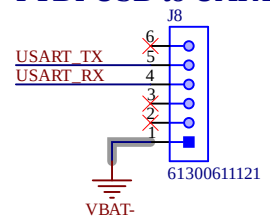


▲ UART will be used to interface with the main control board, sending battery info & state and receiving any commands from the main control board/user

SPI Debug

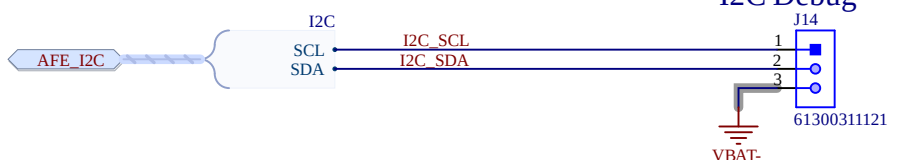


FTDI USB to UART Debug



▲ Breakout headers for debugging/testing

I2C Debug



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PROJECT LTO 12S BMS.PrjPcb, [No Variations]		
DOCUMENT Pack_Interface.SchDoc		MODIFIED 2026-05-18
ENGINEER Farris Matar	REVIEWER *	SHEET 8 OF 7